

Transferable Force Field for Gallium Nitride Crystal Growth from the Melt Using On-The-Fly Active Learning

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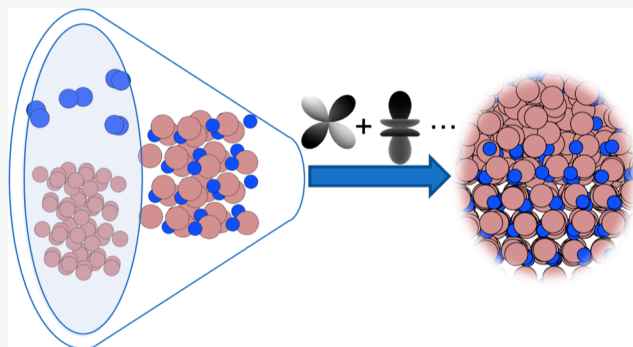


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ABSTRACT: Atomic-scale simulations of reactive processes have been stymied by two factors: the lack of a suitable semiempirical force field on one hand and the impractically large computational burden of using ab initio molecular dynamics on the other hand. In this paper, we use an “on-the-fly” active learning technique to develop a nonparameterized force field that, in essence, exhibits the accuracy of density functional theory and the speed of a classical molecular dynamics simulation. We developed a force field capable of capturing the crystallization of gallium nitride (GaN) during a novel additive manufacturing process featuring the reaction of liquid Ga and gaseous nitrogen precursors to grow crystalline GaN thin films. We show that this machine learning model is capable of producing a single force field that can model solid, liquid, and gas phases involved in the process. We verified our computational predictions against a range of experimental measurements relevant to each phase and against ab initio calculations, showing that this nonparametric force field produces properties with excellent accuracy as well as exhibits computationally tractable efficiency. The force field is capable of allowing us to simulate the solid–liquid coexistence interface and the crystallization of GaN from the melt. The development of this transferable force field opens the opportunity to simulate the liquid-phase epitaxial growth more accurately than before to analyze reaction and diffusion processes and ultimately to establish a growth model of the additive manufacturing process to create the gallium nitride thin films.



1. INTRODUCTION

Reactive–diffusive processes pose a problem for atomic-scale modeling to simulate. The most obvious route for such a study would involve conducting classical molecular dynamics (MD) simulations, which, in turn, would require knowledge of a suitable reactive force field. Many commonly used force fields, such as the modified embedded atom model¹ or Stillinger–Weber² potentials, cannot capture bond-making and bond-breaking processes nor any associated phase change. The well-known Tersoff model³ is a “bond order”-based method and might be considered to offer a compromise in lieu of explicit bond-making and bond-breaking capability, but the limitations of this model have also become well-documented.^{4–6} A more flexible model, such as ReaxFF, which is explicitly designed to capture reactive systems, is certainly an alternative. In practice, ReaxFF requires extensive parametrization, and the result is highly specific to one particular reactive system.^{7,8} If models for the system of study exists, then this is a viable option. The use of an ab initio model, such as density functional theory (DFT) and ab initio MD (AIMD), offers more accuracy but is impractical due to the orders of magnitude with larger computational burden. The cost leads to compromise in accessible length and time scales (typically to a few nanometers in length and order of tens of ps in time for AIMD), without excessive resource usage. Despite these challenges, if a

computational model can be found to capture the reaction, it can expose previously unknown reaction mechanisms that are either inaccessible experimentally or only inferred indirectly, as it is quite common.

Our goal was to develop force fields that exhibit the generalizability needed to model each component over the wide range of temperatures accessible to the liquid-phase epitaxial growth of gallium nitride during an additive manufacturing (AM) process.⁹ The AM approaches allow the fabrication of geometries unattainable through traditional manufacturing and lead to a faster and cheaper prototyping to create a product with targeted material properties.^{10–12} Our collaborators, Gagnon et al., developed an AM method to synthesize the gallium nitride.¹³ This process allows epitaxial growth of crystalline GaN and provides a new route to printing semiconductor materials, including vertical device geometries that enable higher performance but are currently difficult to

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fabricate.¹⁴ However, some questions remained unanswered or inaccessible experimentally, including the inability to resolve the atomic-level details of crystallization at the growing interface, e.g., preferred directions for faceted growth. Nor was there an understanding of the processes by which the nitrogen diffuses through the gallium melt and reacts, again no description at a molecular level. In addition, the AM growth model used by Gagnon et al. underestimates the observed growth rate for unknown mechanistic reasons.¹³

Since no semiempirical model currently fits this requirement, we turned to the use of machine learning to create a force field that can realistically model GaN's crystal growth process. The incorporation of machine learning has advanced our ability to learn the fundamental interactions among atoms and molecules, creating a host of methods such as the Gaussian approximation potential,¹⁵ moment tensor potential,¹⁶ DeepMD,¹⁷ and many more. Recently, researchers have developed a Gaussian process (GP)-based machine learning force field, Fast Learning of Atomistic Rare Events (FLARE).¹⁸ We chose to use FLARE for our purposes because it is easily interpretable and autonomous with little supervision needed. This method falls into the category of an "on-the-fly" (OTF) approach in which more training points are added, usually from a more accurate method (and hence more expensive) only when necessary.^{19–21} FLARE uses the neighboring atoms of a central atom as its descriptor in concert with a GP that makes a prediction of the atomic forces by calculating between the new atomic environment and the ones in the training set. It has the added attractive feature that it compares the uncertainty of a new prediction with the systematic uncertainty (formally an epistemic uncertainty) in order to determine whether the model requires more training points. Compared to a neural network (NN) method, FLARE can start training with a small training set without needing to know the exact reaction pathway or all possible configurations of the system. However, an NN-based method requires a large amount of DFT trajectories, which requires more domain knowledge beforehand instead of actively learning the forces along the way. An NN approach, with its requirement of a large amount of DFT data, leads to an increased computational cost. These two features make FLARE a powerful tool to "learn" the interactions that define the behavior of a complex system incorporating rare atomic events in an "OTF" fashion.

In this work, we used FLARE to develop nonparametric (equationless) force fields that represent both crystalline and liquid GaN as well as the precursors for growth from the melt, namely, nitrogen species diffusing through liquid Ga. Our paper will demonstrate that the OTF training of a GP-based force field showed excellent accuracy and generalizability. During this study, FLARE updated its descriptor and kernel methods and offered a Python application programming interface to a more efficient C++ algorithm.²² This significant change to FLARE allowed us to demonstrate the strengths and limitations of two versions of FLARE (referred to as FLARE 1.0 and FLARE++ throughout this paper) when applied in practice to nonequilibrium, multiphase phenomena involved in GaN crystallization from liquid Ga. We showed that FLARE++ is more powerful and efficient in learning different chemical systems and creating a truly transferrable force field because of its adoption of the polynomial expansion-based atomic descriptors allowing higher-order n -body interactions as well as the implementation of C++ code which allowed a faster computation. To broaden the use of our resulting non-

parameterized, FLARE-generated force field, we "mapped" (to follow the FLARE nomenclature) the GP-generated models onto the large-scale atomic/molecular massively parallel simulator (LAMMPS)-compatible models and parameters.²³ This "mapping" does not manipulate or truncate any existing interactions that arose from the FLARE-generated model. We provide the force field file together with associated parameters for the MD simulations in the Data Availability and [Supporting Information](#).^{22,24} This will enable users in the community to conduct large-scale MD simulations using the nonparametric force field we developed with FLARE but in the more commonly used format of a LAMMPS model.

2. COMPUTATIONAL METHODS

This section covers: (1) the methodology used to create the GP-based force field model through the OTF training and from ab initio MD trajectories and (2) the process used to validate the FLARE force field against established properties in GaN. A graphical schematic of the workflow for the FLARE OTF training is shown in [Figure 1](#). Here, we provide a brief

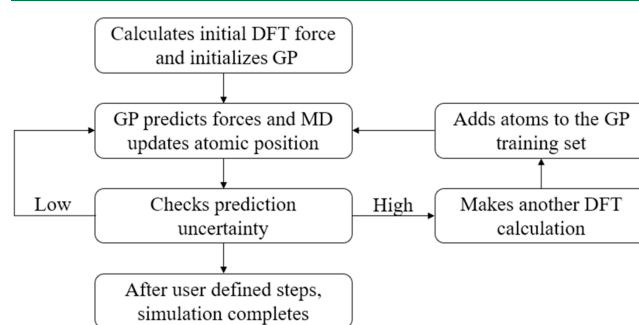


Figure 1. Workflow diagram for FLARE during the OTF training.

overview of the FLARE method, the OTF learning, and the GP-mapped force fields. Further details can be found in the [Supporting Information](#) and in the original papers describing FLARE by Vandermause et al.^{18,22}

2.1. OTF Training of Atomic Systems. FLARE assumes that the local energy contribution to the center atom is the sum of the interactions from neighboring atoms within a user-specified cutoff, r_{cut} .^{25,26} FLARE 1.0 uses an explicit n -body interaction as its atomic descriptor. The kernel is defined as a squared exponential function multiplied by a quadratic function in order to ensure a smooth transition in energy and, critically, in the force at distances beyond r_{cut} as defined in [eq 1](#). $\sigma_{2,3}$ and $l_{2,3}$ are the signal variance and signal length of the 2- or 3-body kernel in the GP. d_i is the vector of the interatomic distance. For the 2-body kernel, $d_i = \vec{r}_{i,i_1}$, where $\vec{r}_{i,i_1}$ represents the distance between central atom i and nearby atom i_1 . For the 3-body kernel, $d_i = (\vec{r}_{i,i_1}, \vec{r}_{i,i_2}, \vec{r}_{i_1,i_2})$.

$$k_{2,3}(d_i, d_j) = \sigma_{2,3}^2 \exp\left(-\frac{\|d_i - d_j\|^2}{2l_{2,3}^2}\right) f_{\text{cut}}(d_i, d_j) \quad (1)$$

The newer version, FLARE++, utilizes the atomic cluster expansion (ACE) descriptor,²⁷ which has three functions of interatomic distance $\vec{r}_{ij}$ as defined in [eq 2](#). The first component, $R_n(\vec{r}_{ij})$, is a radial basis function, a Chebyshev polynomial of the first kind, and $\vec{r}_{ij} = \vec{r}_{ij}/r_{\text{cut}}$. The second component, Y_{lm} , is a spherical harmonic, where l and m are the degree and order of

the functions. The third component is the cutoff function, allowing a smooth transition to the cutoff distance, r_{cut} . Lastly, FLARE++ implements the normalized dot product kernel raised to the power of ξ to compare the similarity between two local atomic environments, $\vec{d}_1, \vec{d}_2$, defined in eq 3 where σ is the signal variance in the local energy and $\vec{d}_i$ is the sum of expanded basis set and spherical harmonics functions in the descriptor. Parameters N (number of the radial basis sets), l , r_{cut} , and ξ are tunable; we discuss a limited search in the Supporting Information.

$$\phi(\vec{r}_{ij}) = R_n(r_{ij})Y_{lm}(\vec{r}_{ij})c(\vec{r}_{ij}) \quad (2)$$

$$k(d_1, d_2) = \sigma^2 \left(\frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|} \right)^\xi \quad (3)$$

FLARE's OTF "learning" is achieved by comparing the inherent GP noise uncertainty, σ_{GP} , and the force prediction uncertainty, σ_{predict} . If σ_{predict} is higher than σ_{GP} , then FLARE automatically makes a DFT call to add more data to the training set. Otherwise, FLARE proceeds with a MD simulation step. During the OTF training phase, we can adjust the temperature so that the system can explore different conditions or chemical phases of the system.

2.2. DFT and Ab Initio MD. We used Quantum Espresso (QE)²⁸ to provide plane-wave DFT calculations. We used projector augmented-wave pseudopotentials with a Perdew–Burke–Ernzerhof exchange–correlation function.^{28–30} The kinetic energy cutoff for the wave functions is 50 Ry, and we used a k -point mesh density of $2 \times 2 \times 2$. Before conducting the OTF training of GaN in a liquid phase, we also used trajectories generated from an ab initio molecular dynamics (AIMD) simulation to "pretrain" the GP model. We found that this was a useful way to generate a large number of accurate DFT "frames" (representing a given time instant) so that the GP model could have a richer understanding of the atomic environment right from the start. The use of AIMD simulations allows us to obtain training data from an accurate (DFT) source but at finite (and experimentally realistic) temperatures. We observed that the use of AIMD frames enabled us to accelerate the OTF training, especially for the larger-sized systems and for those incorporating diffusive and reactive situations.

We used Born–Oppenheimer molecular dynamics (BOMD) as implemented in QE³¹ to generate trajectories and atomic forces at a series of desired temperatures. During the BOMD simulation, QE implements a Verlet algorithm to calculate the equations of motion. We ran different simulations in which we controlled only the initial temperature. We observed that it is best to use frames that are at least 0.1 ps apart from each other to avoid sampling very similar atomic configurations in the training set. We used 80% of the sampled frames for the training and reserved 20% for the subsequent verification of the model against experimental property values.

2.3. Mapping the GP Force Field in LAMMPS. The Gaussian kernel can be mapped onto a low-dimensional grid to represent each 2- and 3-body interaction specified by the user and to maintain the high accuracy of the forces determined from the DFT calculations.²⁴ FLARE 1.0 implements this feature to generate a mapped Gaussian process (MGP) force field compatible with LAMMPS.^{23,24} We mapped atomic models trained with FLARE 1.0 onto a 128-point grid for the

2-body interactions and a $64 \times 64 \times 64$ grid point for the 3-body interactions. The upper cutoff of each grid point is the same as the corresponding Gaussian kernel's upper cutoff. The lower cutoff is defined as the minimum interatomic distance in the training set minus an additional "relaxation" distance (chosen as 0.5 Å in our work). The MGP parameters are interpolated to the lower cutoff so that the system would not fail to converge when the interatomic distance is smaller than the ones in the training set during the LAMMPS simulation. The extra "relaxation" distance is set as a safeguard, and it is always more desirable to include these configurations in the training set. FLARE++ maps the normalized dot product kernel of the GP to a polynomial form. The training points in the GP model can be represented by a symmetric tensor B with rank of the kernel power ξ . As a result, prediction of new atomic environments does not need to iterate through every single point in the training set, and the speed of the FLARE force field is independent of the number of points in the training set.

In contrast to the severe system size constraints experienced in practice using AIMD, the use of FLARE force field allows large-scale MD simulations to be conducted in LAMMPS. For example, in this work we were able to consider the dynamics of more than 10,000 atoms over nanoseconds of simulation, which opens new vistas of processes, like crystal growth, to accurate simulation models. This is achieved by building a bridge between accurate atomic-scale trajectories from GP learning and a more traditional MD simulation.

3. RESULTS

Our task is to use FLARE++ to create individual atomic models for (a) liquid gallium, (b) gaseous nitrogen, and (c) solid gallium nitride as well as, importantly, to create a combined single force field that can capture all three chemical components. We will show how we combined three individually trained atomic models into a single comprehensive force field and tested it in all the phases of matter that are involved in this AM process (solid, liquid, and gas), observing across a large temperature range and validating its prediction of properties relevant to the liquid-phase epitaxy. Finally, we describe the LAMMPS-compatible model that we generated as a result of the FLARE calculations. Later in this section, we discuss the details of OTF training of individual chemical systems using both FLARE 1.0 and FLARE++ and provide details of the training techniques we implemented that improved FLARE's learning process. The details on the determination of the model parameters are included in the Supporting Information.

3.1. Nitrogen. Experimentally, ammonia was used as the source of N atoms in the system. However, ammonia molecules decompose around 1000 K³² well below the AM processing temperature. As a result, we need only to train the atomic model on nitrogen at temperatures between 1000 and 3000 K. Training of nitrogen was crucial to demonstrate that FLARE++ has the ability to model both intra- and intermolecular interactions of nitrogen species. We created a system containing 27 nitrogen molecules, as shown in Figure S3, and carried out the training with Nosé–Hoover ensemble across the temperature range, 300–3000 K. We report the training steps and the corresponding mean absolute errors (MAEs) in Figure S4. The GP model gave a good representation of interactions in the nitrogen system as

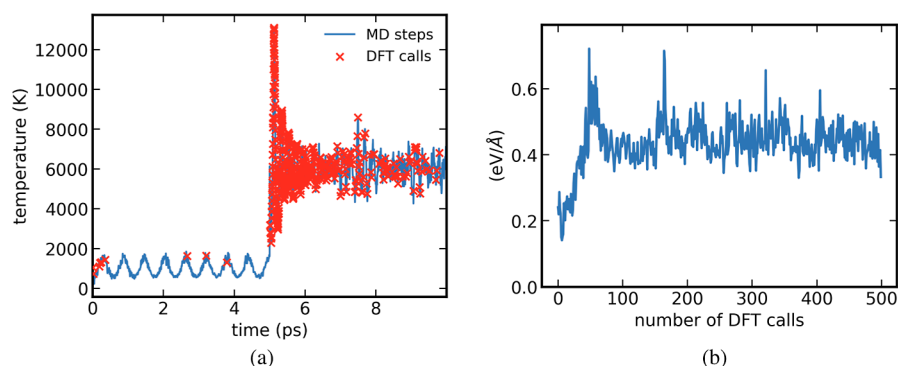


Figure 2. Results from the OTF training steps and the MAE of a larger 64-atom GaN system under a Nosé–Hoover thermostat from 1000 to 6000 K using FLARE++. A total of 11 DFT calls were made at 1000 K and 491 DFT calls at 6000 K. DFT calls shown as red crosses in (a). The MAE shown in (b) is 0.16 eV/Å in the solid phase and 0.40 eV/Å in the liquid phase. (a) OTF training of temperature versus time. (b) Corresponding MAE values.

exemplified by the low MAE value of forces around 0.05–0.08 eV/Å.

3.2. Gallium. Liquid gallium serves as a critical reactant in the liquid-phase epitaxy to produce crystalline GaN. Gallium has a low-melting temperature of 302.92 K.³³ During GaN nucleation and crystal growth, liquid gallium serves as a “reservoir” for the reaction to take place. Our active learning approach must then be able to capture the behavior of the liquid gallium metal. To do so, we set up an OTF training of 64 gallium atoms in a simulation cell with periodic boundary conditions using FLARE++, as shown in Figure S1. We ran the MD training under isobaric–isothermal (NPT) conditions at 1 bar for 10 ps at two different temperatures, 100 and 2000 K. The MAE between the atomic forces from DFT calculations and those of the GP prediction was around 0.1 eV/Å with only 46 DFT calls, as shown in Figure S2. Typically, we aim to achieve an MAE of less than 0.2 eV/Å or ~10% of the absolute atomic forces.

3.3. Gallium Nitride. Benefiting from the more efficient kernel method and algorithm, the OTF training of FLARE++ is fairly straightforward. We used the 64-atom GaN system with a Nosé–Hoover thermostat for the OTF training. The training started at 1000 K for 5 ps and adjusted to 6000 K for another 5 ps. The training profile and the corresponding MAE are shown in Figure 2. The model needed little effort to learn the low-temperature, crystalline region with just a few DFT calls. The MAE of atomic forces in the first 5 ps was 0.16 eV/Å with 11 DFT calls. Once the system temperature increased to 6000 K, more data points were needed. The MAE increased to 0.4 eV/Å with 491 DFT calls after the temperature increased to 6000 K and the system entered the liquid phase. We believe that two factors contributed to this relatively high error. First, it was harder to predict the atomic forces in the more disordered liquid phase at high temperature. Second, at higher temperature, the magnitude of the atomic forces as well as the error increased correspondingly. This occurrence was also recorded in the original FLARE paper that studied the melting of Al.¹⁸ In our GaN system, the average force component above the melting temperature was about three times higher than ones recorded below the melting temperature. We calculated the percentage error of the force prediction in the liquid phase at higher temperatures to be around 15% of the force magnitude, which is consistent with values at lower temperatures. As a result, we believe that increases in the MAE should not be used as the sole metric to determine the GP’s

capability to model complex systems. Instead, it is prudent to consider the increase in the system temperature and compare the percentage error of the force prediction and the magnitude of the forces. The OTF training profile and its corresponding MAE error are shown in Figure 2. The atomic forces, total energy, and the system stress tensor were used as “ground truth” labels during these trainings.

3.4. Combination of Atomic Models. We used two protocols to combine the existing atomic models for vapor, liquid, and crystal systems. One method was to simply concatenate the local atomic environments within the training sets from multiple OTF-trained models by merging those environments into a single atomic model. Another method is to conduct additional OTF training on the DFT-calculated trajectories invoked during the training of each chemical system. Both methods have their advantages and limitations. The first method guarantees the sampling of each of the chemical systems that the original atomic models represent. However, the size of the training set could become large as a result and make hyperparameter optimization memory intensive. The second method is computationally efficient because it samples only atomic environments with high uncertainty. However, this automated process might oversample such data points, resulting in an unwanted bias. For example, the algorithm could keep picking atomic configurations from gallium nitride in the liquid phase and ignore ones from liquid gallium, causing inaccurate density calculations during the gallium melting process.

Here, we combined atomic models trained independently from gallium, gaseous nitrogen, and gallium nitride, as presented above. We also tested the model accuracy of both combination methods and present their performance in the S. The first method worked better in our case because it can more accurately model the gallium melting and nitrogen diffusion than the second method. We used the normalized dot product with a power of 2 as the kernel method. The hyperparameters of the final comprehensive atomic model are as follows: the cutoff distance is 4.0 Å, the number of radial functions is 8, and the number of spherical harmonics is 4. The optimized model has an energy signal variance of 4.61 eV, an energy noise of 0.37 eV, a force noise of 0.08 eV/Å, and a stress noise of 0.24 GPa. The training set consisted of 501 configurations from gallium nitride, 46 configurations from gallium, and 82 configurations from gaseous nitrogen. Details on our search for hyperparameters are included in the

Supporting Information. We compared the atomic force and energy results between DFT calculations and our force field's prediction on both the training set and an independently calculated test set in the Supporting Information as a direct evaluation.

3.5. Creating a LAMMPS-Compatible Model and Property Validation. After the training, we mapped the FLARE++ model to an LAMMPS-compatible “pair style” in order to make the learned GaN model parameters more widely accessible to users. To validate the LAMMPS-compatible force field against known experimental data, we calculated the melting temperatures of gallium and gallium nitride, the diffusion coefficients of nitrogen at different temperatures, and the lattice parameter and elastic tensor of gallium nitride crystals. In these MD simulations, we used a time step of 1 fs. The melting temperature of gallium was estimated to be 308 K and the density increased by 5% upon melting with a density maximum at 5.83 g/cm³. Comparisons to their experimental counterparts are described below. To calculate the self-diffusivity of nitrogen, we equilibrated the simulation box in an isobaric–isothermal NPT ensemble at atmospheric pressure followed by an isothermal NVT ensemble using a Nosé–Hoover thermostat for 5 ns in total. We then ran the microcanonical NVE ensemble for a further 5 ns until the mean squared displacement data versus time entered a linear region, as recommended in previous literature.³⁴ The nitrogen diffusivity was estimated to be 0.71 cm²/s at 300 K and atmospheric pressure in reasonable enough agreement with an experimental value of 0.212 cm²/s as simulated diffusivity could overestimate by 1–9 times of experimental value.^{35–38} The FLARE++ model was able to capture an expected Arrhenius relation between diffusivity and temperature from our extended temperature range from 300 to 2000 K, as shown in Figure 3. Although diffusion coefficients above 1200 K

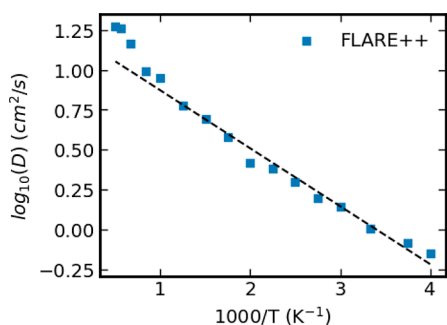


Figure 3. FLARE++-calculated values of the self-diffusivity of nitrogen, showing the expected roughly linear relationship between the inverse of the temperature (K^{-1}) and the natural logarithm of the diffusion coefficient over a very broad range of temperatures.

(around 0.8 on the x -axis) deviate above the linear relation, our single transferable force field is still capable of capturing diffusivity in a gaseous phase over a very large range of temperature.

To calculate the elastic constants of GaN, we calculated the derivative of the stress divided by the stress tensor. We applied a 0.0001% positive deformation to each surface of wurtzite supercell and observed that the elastic constant remained the same over a range of deformations from 0.0001 to 1%.²³

The efficacy of the LAMMPS model to predict system properties is presented in Tables 1 and 2, where this is compared to values calculated via independent gallium nitride

Table 1. Lattice Parameters of a Wurtzite Structure of GaN Predicted Using the FLARE 1.0 and FLARE++ Force Fields Compared to Known Values from DFT Calculations⁴⁰ and Experimental Measurements^{42a}

lattice (unit)	FLARE	FLARE ++	combined model using FLARE++	DFT (GGA)	DFT (LDA)	expt.
a, b (Å)	3.26	3.19	3.22	3.20	3.07	3.22
c (Å)	5.20	5.22	5.25	5.23	4.91	5.24
maximum deviation (%)	1.2	0.9	0.2	0.6	6.3	N/A

^aAll values are rounded to the second decimal place.

Table 2. Calculated Elastic Constants Using FLARE 1.0, FLARE++, and the Combined Force Field Compared to Known Values from DFT Calculations³⁹ and Experimental Measurements⁴¹

C_{ij} (GPa)	FLARE 1.0	FLARE ++	combined model using FLARE++	DFT (GGA)	DFT (LDA)	expt.
C_{11}, C_{22}	272	527	456	354	374	390 ± 15
C_{12}	130	246	195	83	127	145 ± 20
C_{13}, C_{23}	96	198	156	52	81	106 ± 20
C_{33}	221	454	401	380	442	398 ± 20
C_{44}, C_{55}	49	123	114	98	99	105 ± 10
C_{66}	71	140	130	136	124	123 ± 10
RMSE	65	59	28	27	16	N/A

models trained using both FLARE 1.0 and FLARE++, as well as experimental values and those from DFT calculations found in the literatures.^{39–42} Although the FLARE++ force fields were not explicitly parametrized to fit the lattice parameter or mechanical properties, we observed that it is capable of representing experimental behavior with remarkably high accuracy in comparison to both DFT and experimental values. Using the recombined model, our calculated lattice parameters showed a maximum deviation of approximately 0.2% from the experimental values. The overall root-mean-square error (RMSE) of the elastic tensor estimation is 28 GPa. Compared to the RMSE arising from two DFT calculations of 27 GPa (GGA) and 16 GPa (LDA),³⁹ our FLARE++ force field was as quantitatively accurate as DFT to reproduce experimental values. The calculated elastic tensor was slightly overestimated compared with the experimental value. The ability of the FLARE++ force field to generalize its predictive property generation seems promising and could allow the potential investigation of mechanical properties in nanomaterials without exhaustive parameter fitting.

Estimation of the equilibrium melting point required us to deploy a more sophisticated method. Indeed, the melting GaN is challenging even in experiments because GaN shows a pronounced tendency to decompose to Ga and N₂ before reaching the melting point.⁴³ Utsumi et al.⁴³ reported a successful melting point measurement of 2488 K at 6 GPa that avoids the decomposition observed at lower pressures. If we were to perform the MD simulations of a bulk crystal of GaN at successively increasing temperatures in the vicinity of the experimental value of the congruent melting point, it is well known that the materials tend to overheat up to the spinodal temperature, thus overestimating the melting point. Overestimates for other semiconductors can be as much as 40%

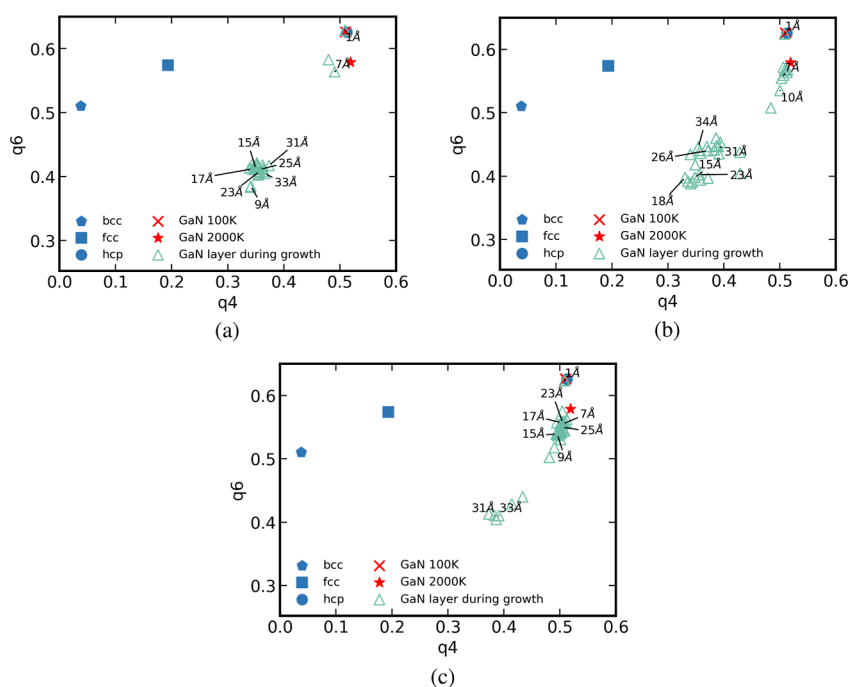


Figure 4. Steinhardt order parameters q_4 and q_6 are grouped in the 2 Å wide layers during the GaN crystal growth simulations. The order parameters from each layer are marked as triangles. We annotated every other layer with its distance from the center of the simulation box. The square, pentagon, and circle symbols represent fcc, bcc, and hcp reference structures, respectively. The red cross represents the wz-GaN crystal at 100 K, which overlaps with the circle of the hcp reference structure. The red star represents the wz-GaN crystal at 2000 K overlapping with the triangle symbol of the crystallized local hcp structure after our crystal growth simulation. (a) 0, (b) 2, (c) 4 ns.

above the congruent melting point. For example, Zhou et al.⁴⁴ recorded a melting point of 3570 K at zero pressure using the Stillinger–Weber potential for GaN and an overheating of almost 1000 K over the experimental result at 6 GPa. Harafuji et al.⁴⁵ derived a pressure-dependent melting point equation for GaN using a modified 2-body potential. Although their P – T relation produced a melting point of 2538 K at zero pressure and 3433 K at 6 GPa, it failed to take into account the overpressure requirement to achieve melting in the experiment.

In determination of the melting point in this study, we first simulated the behavior of a bulk GaN crystal modeled by simply increasing the temperature in order to determine the limit of mechanical stability of the system (the spinodal), which occurred at around 3200 K at 6 GPa. Next, we set up a simulation box of 6912 atoms in which we created a solid–liquid interface. As is well known, the interface present in this setup prevents the superheating inherent in the first (surfaceless) method.^{44–46} The entire box was relaxed at 2800 K at a pressure of 6 GPa in an isothermal–isobaric NPT ensemble for 1 ns. The ordered atoms in the left half of the box (z -direction) were designated as the wurtzite crystal phase and fixed to maintain the structure. The atoms in the right half of the box were designated as the liquid phase. The atoms in this liquid phase were heated up to 3250 K over the course of 4 ns and then equilibrated for another 4 ns. During the melting process, we used an NPT ensemble to maintain a 6 GPa pressure in the direction of the interface (z -direction); dimensions in the x - and y -directions were held fixed. Lastly, all the constraints were removed, and the system was relaxed in the NVE ensemble for 4 ns. The time step during this entire process was 1 fs. The determination of the melting point, T_m , was made as follows: We started from the superheated melting point (3250 K) and decreased the temperature of the entire

system in increments of 100 K. We monitored the density profile and Ga–N coordination number along the z -dimension after the final NVE simulation as shown in Figure S11. The temperature was lowered until the point at which the density and coordination number profile showed equal volumes of the solid–liquid region and the velocity of the moving solid–liquid interface was zero. Our prediction of the melting point was 3067 K at 6 GPa. The machine learned force field offers the closest prediction of T_m to the experimental value (ours overestimates by 24%) compared to previous studies that overestimate by 40% or more.^{44,45}

3.6. Crystal Growth of Wurtzite GaN. Lastly, we showed that the FLARE++ force field is capable of capturing the formation of the local hexagonal close-packed (hcp) structure of the wurtzite (wz) GaN as a result of the reaction between Ga and N atoms in the melt. We adopted a sandwich-like simulation setup wherein 4 layers of crystalline wz-GaN(001) were located in the middle of the simulation box, sandwiched between slabs of 2000 gallium and 2000 nitrogen atoms, randomly placed on both sides of the crystalline GaN to mimic liquid phases. During the simulation, the innermost two layers of the crystal were fixed in place to maintain their positions in the box as well as to provide a crystal seed for the surrounding disordered, liquid-like atoms to encourage crystallization. The system temperature is an important factor during crystal growth simulations; it defines the undercooling and, hence, the crystallization velocity. Recapping, our force field predicted a melting temperature of around 3067 K at 6 GPa, compared to the experimental melting temperature of 2488 K at 6 GPa.⁴³ We chose to use a temperature of 2000 K, i.e., well below both of these values of T_m to simulate crystal growth under a constant-temperature NVT ensemble for 4 ns.

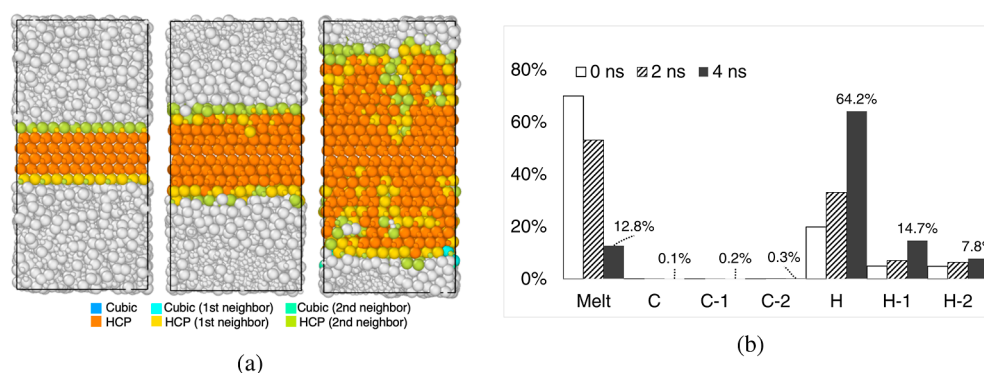


Figure 5. (a) Visualization of a few cubic and prevalent hexagonal structures during gallium nitride crystal growth at three MD snapshots corresponding to 0, 2, and 4 ns time-points. Color code shown below the snapshots. (b) For the same three time-points, the percentage of atoms that belong to the disordered melt region (steadily decreasing over time), cubic (C), cubic arrangements among the first-nearest neighbors (C-1), cubic arrangements among the second-nearest neighbors (C-2), showing very little cubic arrangements, and hexagonal (H), hexagonal arrangements among the first-nearest neighbors (H-1), and hexagonal arrangements among the second-nearest neighbors (H-2), which grow steadily over time.

We calculated the widely used Steinhardt order parameter for all of the atoms in the system to verify the local crystal geometry.^{47,48} Following the order parameters as a function of simulation time allowed us to track the extent of ordering in the system. Details on Steinhardt order parameter calculations and the characterization procedure are included in [Supporting Information](#). In crystallography, q_4 and q_6 are used together to examine the local structural geometry. To determine which structure is forming at the crystalline wz-GaN surface during our simulation, we calculated the order parameters characteristic of face-centered cubic (fcc), body-centered cubic (bcc), and hexagonal close-packed (hcp) lattices as reference points. We also calculated the order parameters of bulk wz-GaN at both 100 and 2000 K. We grouped atom-wise q_4 and q_6 into the 2 Å wide bins in the direction perpendicular to the GaN surface and averaged the order parameters in each bin. This allowed us to visualize the development of a local structure in each layer, which we found to resemble a layer-by-layer crystal growth process.

We have plotted the values of q_4 and q_6 from each layer at three different time-points in the simulation: 0, 2, and 4 ns, along with the reference crystal structures in [Figure 4](#). First, the q_4 and q_6 parameters clearly distinguished differences between fcc, bcc, and hcp lattices, as expected. Second, wz-GaN showed a uniquely hcp local structure at 100 K as represented by the overlapping positions of the blue circles and red crosses in [Figure 4](#). The q_6 parameter was 8% lower due to the higher thermal fluctuations that are extant at 2000 K. Lastly, we looked at the time evolution of order among the initially disordered liquid-like atoms in the simulation box. At the start of the simulation (0 ns), we confirmed that these atoms did not belong to any reference crystal structure. However, once the simulation started, layers in the disordered regions gradually began to crystallize and formed a characteristically local hcp structure. Unsurprisingly, the atoms closer to the wz-GaN surface crystallized first. After 4 ns of simulation, all the layers except the ones furthest away from the substrate can be seen to have formed an hcp structure; their q_4 and q_6 values converged near the order parameter of the bulk GaN crystal at 2000 K, represented by a red star.

We visualized the crystal growth and identified the cubic and hexagonal structures with OVITO as shown in [Figure 5](#).^{49,50} The percentage of atoms crystallized into hexagonal structures

was 86.7%; in contrast, the cubic structures constituted less than 0.5% throughout the simulation. We did not observe a large number of defects or in-plane grain boundary due to fact that, first, we used a system size that is orders of magnitude smaller than an experimental scale, and hence, we should not expect to find many defects. Second, the substrate we used was wurtzite GaN *c*-plane, so there is little lattice mismatch compared to growing on a foreign substrate such as sapphire or silicon carbide. The Steinhardt order parameter and the diamond structure characterization showed that the FLARE++ force field accurately represented the crystal growth of wz-GaN from a region containing the disordered gallium and nitrogen atoms.

3.7. Improvements to the Training Process of the FLARE Versions. Now, we revisit the training and validation of individual atomic systems with both FLARE 1.0 and FLARE ++ in light of our experiences with this study. FLARE 1.0 is implemented in Python, which does not perform well with memory-heavy, parallel codes, and the combination process became extremely slow after around 2000 data points had been sampled. Indeed, the process became essentially stagnant after 3000 data points had been sampled. The MD simulation below used a force field trained specifically for each chemical system. However, FLARE++ is written in C++ to address the computational bottleneck, which enabled the construction of large-scale atomic models. The next paragraph reports the results of MD simulations used with FLARE++ to show the combined model of the three chemical systems in order to demonstrate the transferability and accuracy of the force field that it creates to investigate materials properties.

An LAMMPS simulation using the combined FLARE++ force field successfully captured the liquid-phase behavior of gallium including its melting, which allowed experimental verification. We used an NPT ensemble to simulate a supercell of 4096 gallium atoms at temperatures from 100 to 1000 K at a pressure of 1 bar with a 1 fs time step. We increased the temperature in increments of 50 K and allowed the contents of the simulation box to relax for 1 ns at each temperature. We estimated the melting point for gallium to be 310 K, which is in good agreement with the experimental melting point of α -gallium (302.92 K).³³

The density profiles over this temperature range and the radial distribution function, $g(r)$, are shown in [Figure 6](#).

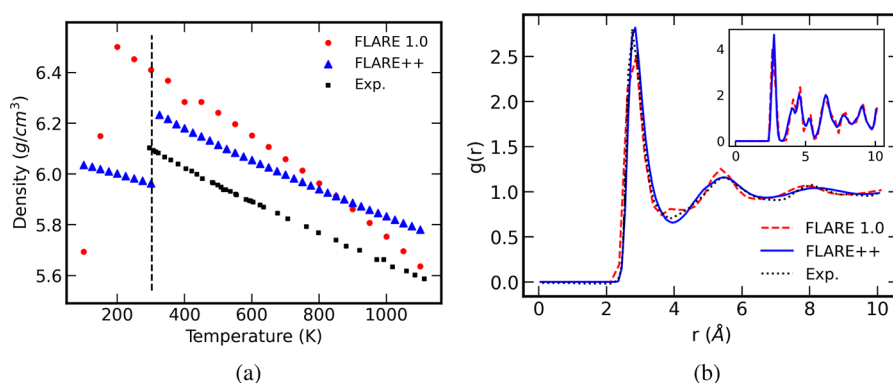


Figure 6. Predictions of density values and radial distribution functions using FLARE 1.0 and the combined FLARE++ force field. (a) Comparison of the density profiles of a simulated Ga supercell using FLARE 1.0 (red dots) and the combined FLARE++ (blue triangles) and experimental data (black squares).⁵⁶ The vertical dashed line shows the location of the experimental melting temperature of 302.92 K.³³ Errors of the simulated density values were below 1%. (b) Average radial distribution function, $g(r)$, of a Ga supercell of a solid phase at 100 K (inset) and a liquid phase at 500 K predicted by the simulation after 1 ns of relaxation. $g(r)$ simulated using FLARE 1.0 is shown as a red dashed line. That simulated by the combined FLARE++ approach is shown as a solid blue line. The experimental measurements of $g(r)$ in a gallium liquid are shown as a black dotted line.⁵⁹ The expected transition from crystal to liquid is clearly evident. The close comparison of experimental and FLARE-predicted $g(r)$ values for liquid Ga is also apparent.

Gallium is known to have several distinct solid phases with densities ranging between 5.91 and 6.57 g/cm³.^{51–55} Our FLARE++-generated force field overestimated the α -gallium density by just 3% across the entire simulated temperature range, 100 to 1000 K. Like water, gallium exhibits a density maximum on melting, with the density of liquid gallium being higher than that of solid gallium by up to 3%. The density has been shown experimentally to decrease once the gallium enters the liquid phase.⁵⁶ Our force field also showed a density increase of 4.5% upon melting, with the density maximum located at around 320 K.

A similar study using FLARE 1.0 was less successful. Although the MAE during FLARE 1.0 training was below our target of 0.2 eV/Å, the subsequent LAMMPS simulation resulted in an “explosion” of atoms in the simulation cell. We believe this was caused by strong repulsive forces at the small interatomic distances between gallium atoms during melting and the limitation of the 2- and 3-body descriptors to represent the increased randomness characteristic of the metallic liquid. We remediated this situation by starting the OTF training with a compressed lattice with a 10% decrease in the initial Ga–Ga distance. This forced the model to, in some way, “understand” what should happen when gallium melts. Our modification allowed the FLARE 1.0-generated force field to capture the melting process. However, it underestimates the melting temperature by about 200 K and overestimates the density by at least 5% at lower temperatures from 100 to 600 K.

We used the visual MD program⁵⁷ to calculate the radial distribution function, $g(r)$, of solid and liquid gallium simulated with FLARE 1.0 and the combined FLARE++ model which showed similar results. At 200 K, the first peak in the simulated $g(r)$ appeared at a distance of 2.75 Å, representing the nearest neighbor distance in the gallium crystal. This is consistent with the experimental measurements varying between 2.48 and 2.79 Å.⁵⁸ The second peak in our simulations occurred at 4.15 Å, which is perhaps acceptably close to the experimental measurement at 3.75 Å.⁵⁸ The change of $g(r)$ from 200 to 500 K shows a transition from solid to liquid phases; the magnitude of the first peak decreased and the second and subsequent peaks broadened and lowered in value. Solid Ga has peaks around 2.4, 4.2, and 6.4 Å, whereas

liquid Ga only has peaks around 2.4 and 5 Å. Our calculated $g(r)$ are consistent with the experimental measurement as shown in Figure 6.^{59,60} Although both models can capture the structure difference in solid and liquid phases, the FLARE++-trained model showed better consistency with experimental values and had better ability to capture the melting behavior of gallium.

For the GP model training of GaN using FLARE 1.0, we implemented several different training procedures to determine the best protocol to follow. We initialized a 32-atom GaN wurtzite structure at room temperature (300 K) and then increased the temperature rapidly every 2 ps to 500, 800, 1000, and 1200 K. We then transferred the training to a larger 64-atom GaN system, as shown in Figure S5. The GP model already had sufficient knowledge of the lower temperature regime. It did not make any additional DFT calls below 1000 K and only needed to make 10 additional calls at higher temperatures to further improve the model. The training statistics in Figure S6 show that the model exhibited excellent performance. This was measured by the maximum MAE value being under 0.1 eV/Å and around 10% of the magnitude of the force, even using a relatively large uncertainty threshold of twice the epistemic uncertainty.

When we continued the training at temperatures above 1200 K, we encountered a significant slowdown in the execution of the MD steps. We hypothesized that this phenomenon is caused by two factors: First, the GP model already contained many training points (more than 1000 atomic environments if we started from 300 K) and the inference of the GP model given new atomic coordinates would require more computational time. Second, experiments have shown that GaN tends to decompose prior to melting, and the experimental melting point of the GaN is only known to be the neighborhood of 2500 K at a pressure of 6 GPa.⁴³ In our simulations, it is possible that GaN underwent decomposition and a phase change when the system temperature approached the melting point. This may be the reason the OTF model needed to add significantly more DFT data points to the training set, slowing the process significantly and worsening the GP’s time performance, as pointed out in the first factor. To remedy the slowdown, we could limit the number of atomic

environments added after each DFT call or reduce the maximum number of times the GP is allowed to optimize its hyperparameters. Both of these processes would accelerate the execution speed of the model. However, this might endanger the accuracy of the GP model by setting strict limitations on the training set size or optimization steps.

Instead of choosing either of those remedies, we decided to run the AIMD simulations at 1000, 1500, and 4000 K, which allowed us to sample atomic coordinates and forces at higher temperature regions which were hard to reach, or maintain, in the OTF training. We made sure the atomic environments in the pretrained model did not exceed 500 points and its MAE on the test set was 0.09 eV/Å for frames from 300 to 1000 K and 0.17 eV/Å for frames from 1000 to 2500 K. The MAE values for the training and test sets are both acceptably low, under 0.2 eV/Å. Details of pretraining using AIMD trajectories are included in the [Supporting Information](#). We then restarted the OTF training of GaN in the 64-atom system and rescaled the temperature to 6000 K every 0.5 ps for a total of 2 ps, as shown in [Figure S8](#). This approach makes sure that the system would be maintained in high-temperature regions and that the GP model could effectively sample atomic configurations from the liquid phase. This tailored control is necessary only when we want to build a force field capable of modeling both the crystal and liquid phases. Similar to the transfer from the 32-atom to 64-atom GaN system discussed above, the transferred model only needed additional DFT calls in the liquid phase when the temperature was above 4000 K. MAE values of the restarted OTF training of the GaN liquid phase lie between 0.3 and 0.4 eV/Å, higher than the previous training statistics, which tended to be under 0.2 eV/Å. The transfer training of the GP from the AIMD trajectories to OTF training allowed us to control the number of training sets in the low-temperature region and allowed the active learning approach to sample more effectively in the liquid phase. We implemented transfer learning of smaller systems and low-temperature trajectories to accelerate the FLARE 1.0 training of GaN to find the optimal between model accuracy and computational efficiency.

4. DISCUSSION

In this work, we have successfully developed a nonparametric and robust OTF force field that enabled us to observe crystallization of gallium nitride taking place from a liquid phase. As a precursor to this, we developed and validated FLARE's ability to model each of the two components (gallium and nitrogen) that must react to form the GaN crystal. We compared FLARE's predictions against experimental data with a single force field that combined information from all of these phases of matter.

We showed that this single FLARE++ force field made accurate predictions of atomic forces in each phase (gas, liquid, and crystalline solid) with mean absolute percentage errors of around 10–15%, which can be used as a convenient metric to monitor the training accuracy. Subsequent validation of the force field using MD simulations (postmodel training) produced properties consistent with experimental results over a broad range of properties, including the radial distribution function, lattice parameters, elastic constants, and determination of the melting point. These results generally outperformed those from traditional force fields with the accuracy of the DFT. Our force field exhibited good generalizability to reproduce different physical properties, importantly without explicitly fitting the model to those properties, in the training

data. This development is significant because it does not solely focus on the training or validation statistics of the model. Instead, we focus on testing the physical properties of gallium nitride and its precursors in larger-scale molecular simulations using the FLARE-generated force field to be used within a LAMMPS framework. This provides options for readers to implement LAMMPS's well-used force field framework for their own studies of the microscopic behavior of gallium nitride under a broad range of temperatures and phases. Most importantly, we demonstrated the successful simulation of wz-GaN crystal growth from the melt using the FLARE++ force field, confirming the local structure by calculating Steinhardt order parameters. The development of this transferable force field will allow us to further investigate the physical behavior of the crystal growth of GaN and eventually guide the process and synthesis control of the AM of other semiconductor materials.

Lastly, we want to acknowledge that the DFT calculations are not spin-polarized during the training because the atomic descriptor itself does not distinguish between spin states. Thus, the training data from spin-polarized calculations and nonspin-polarized calculations could have similar atomic environments. Mixing the two calculations would require FLARE to learn two energy distributions with the same descriptor representation. Also, we have shown that our model is transferable to a high-temperature region where spin-ordering in the electronic degree of freedom would not be prominent, as pointed out by the FLARE developers.⁶¹

The experience we gained during the conduct of this research also allows us to make a number of recommendations that we found improved the training and validation procedures for this OTF model. For example, we used transfer learning to pretrain with the AIMD trajectories for gallium nitride liquid-phase simulations. We introduced lattice compression to the gallium system to cope with the poor representation of decreased interatomic distances during melting. One of the advantages of using FLARE is that the GP model learns OTF without much human intervention. It samples diverse atomic configurations to enrich the training set by exploring different temperature ranges. This is established on the assumption that temperature is a sufficient driving factor to enter a different atomic space (e.g., a remote region on a map that is crucial to describe the overall behavior of the system) instead of an "uncertainty maximum." Additionally, it is likely to be computationally expensive to build a training set by sampling atomic configurations over a large temperature range. It requires additional DFT calculations and will increase the prediction time of the GP as the size of the training set increases. However, we can manually add atomic configurations which the model might fail to predict or start the training with a set of reference configurations, as we did in the gallium training with lattice compression. However, this approach has the downside of defeating the purpose of achieving an unbiased and generalizable model by controlling only the temperature during the training. This is an important trade-off during the active learning process. In our work, we combined individually trained models from each chemical environment involved during the gallium nitride crystal growth process to improve the overall force field representation. In the end, we recommend that users validate their model in MD simulations and compare the results against experimental measurements or simulated properties using DFT or a traditional force field.

Not surprisingly, given that this is the newer of the two versions of FLARE, our experience is that FLARE++ is more powerful than FLARE 1.0 with respect to model training efficiency and materials' property predictions. FLARE++ adopts the atomic cluster expansion descriptor and defines the local atomic interaction with the element-based polynomial functions. Compared to the simple 2- and 3-body descriptors adopted by FLARE 1.0, FLARE++ allows a more efficient implementation of many-body models than simple pairwise and triplet interactions. The ACE descriptor also incorporates the angular contribution of nearby neighbors to the center atom as a density cloud. This allowed the computational cost of FLARE++ to scale linearly with the number of data points sampled during the OTF training, a large improvement over that of FLARE 1.0. FLARE++ further improves the capability of MD force fields to learn how to represent complicated chemical systems and establish a larger and more comprehensive model over multiple phases.

■ ASSOCIATED CONTENT

Data Availability Statement

A file containing information on the LAMMPS-compatible force field can be downloaded at <https://doi.org/10.24435/materialscloud:57-cj>.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.3c00587>.

FLARE OTF training process, hyperparameter optimization, and force field validation details with respect to determination of the elastic constants and the order parameters (PDF)

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Author Contributions

X.C., N.Q.L., and P.C. were engaged in the design of the project. X.C. conducted the computations and analysis. X.C. wrote the manuscript in conjunction with N.Q.L. and P.C. W.S. contributed to the calculation of elastic constants. P.C. supervised and provided critical feedback to the project.

Notes

The authors declare no competing financial interest.

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